Computed Targets as Mathematical Guardrails for LLM Negotiators: Anchoring Defense and Variance Collapse in Multi-Issue Bargaining

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Abstract

We study whether external strategic models can improve LLM negotiation outcomes in multi-issue bilateral bargaining. A NegotiationGRU (11 K parameters) trained on heuristic-agent data computes concrete numeric target proposals injected into the LLM's prompt as hard directives. Our central observational finding is that *text-based strategic advice backfires*—aligned LLMs interpret opponent-preference information as a cooperation signal, consistent with alignment-induced cooperation bias—while computed numeric targets produce two distinct mechanisms plus a formula-convergence property.

Scaled experiments ($N = 200$, randomized opener, Claude 3.5 Haiku) decompose the computed-target effect into: (1) **anchoring defense** (+7.1pp), neutralizing the seller's first-mover advantage; (2) a **residual opponent-modelling edge** (+3.0pp), persisting when the anchoring confound is removed; and (3) **variance collapse** (2.3–9.9×), reducing outcome standard deviation from ±8.9% to ±0.9–3.8% depending on the weight source. As a formula property, an **Oracle ablation** shows that with accurate opponent weights the cooperative blend in Equation (3) mechanically converges toward the Nash bargaining solution—highest Pareto efficiency (79%), tightest variance (±0.9%), lowest buyer share (49.5%). We discuss this as a property of the chosen blend, not a fundamental law (Section 5.3).

Replication across **eight models from five labs**—Claude Haiku, Claude Opus 4.6, GPT-5, Gemini 3 Pro, Gemini 3 Flash, Grok 3, Grok 3 Mini, and DeepSeek R1 ($N = 50$ each, randomized opener)—reveals a **three-tier taxonomy**: (i) *OM-effective* models (Grok 3, Haiku, Grok 3 Mini) exhibit 2.0–2.8× variance collapse and +3–7pp buyer advantage; (ii) *collapse-only* models (GPT-5, Gemini Flash, DeepSeek R1) show 1.4–1.6× variance reduction without surplus shift; and (iii) *no detectable effect at* $N = 50$ (Opus 4.6, Gemini Pro), where baseline variance is already low (±4.5–8.0%) and our test is underpowered to resolve sub-±5pp shifts. The strongest predictor of computed-target effectiveness is baseline negotiation stochasticity rather than model capability.

Cross-model experiments pairing a cheap Haiku buyer (with opponent model) against frontier sellers (without) show the opponent model preserves a +5.2pp edge against Grok 3 but cannot overcome the capability gap against Opus 4.6 (49.1%), GPT-5 (48.1%), or Gemini Pro (47.9%); cross-model variance collapse also disappears (±11.3–20.3%). **Within-model**, computed targets give a $0.25/M-token model the outcome consistency band of frontier models; **across-model**, the cheap-with-OM stack still loses on share against frontier sellers, so the "inference compute arbitrage" we report applies to the consistency dimension only.

Finally, a **neural policy** (6.5K-parameter MLP) trained via behavioral cloning and reward-weighted regression on 660 live LLM games successfully crosses the sim-to-real gap with 100% deal rate. Reward-weighted regression at $\tau = 0.2$ converges to the lowest-variance policy observed (±2.6%). We have not run a τ-sweep, so we present this as a hypothesis

suggested by a single-temperature result—offline RL in stochastic multi-agent environments may favor variance minimization—rather than an established convergence property.

A diagnostic re-analysis (Section 4.10) shows that the trained GRU's weight predictions on LLM trajectories are statistically at chance and its 11,078 parameters can be replaced by a 15-line analytical estimator with one tunable scalar. Variance collapse derives entirely from the directive format; the residual share gain reduces to a bias parameter. We recommend reading the architecture as a structured numeric sub-agent rather than as a learned opponent model.

1 Introduction

Large language models are increasingly deployed as autonomous negotiation agents [Fu et al., 2023, Abdelnabi et al., 2024], and recent at-scale deployments are starting to surface the practical consequences. Anthropic's *Project Deal* [Anthropic, 2026]—a real-money agent-to-agent marketplace with 69 participants and 186 deals—reports two findings that closely parallel ours: (i) instructing agents "to negotiate aggressively had negligible statistically significant effects," echoing our finding that text-based strategic advice backfires, and (ii) users represented by stronger models (Opus) systematically extract more surplus than users with weaker models (Haiku), without the disadvantaged users perceiving the gap (fairness ratings 4.05 vs. 4.06). Their work demonstrates the problem at deployment scale; we contribute the mechanistic decomposition and an architectural remedy.

Their natural-language fluency makes LLMs compelling candidates for multi-issue bargaining, where integrative potential depends on identifying and exploiting asymmetric preferences. Yet language fluency alone does not confer strategic competence. An agent must infer what the opponent values, decide where to concede and where to hold firm, and communicate offers that advance its position—three capabilities with fundamentally different computational requirements.

Prior work on LLM negotiation typically conflates these roles: the same model that generates natural language also performs belief inference and strategic planning. We argue this conflation is a design error when the LLM is RLHF-aligned. Alignment training biases the model toward cooperative, other-regarding behavior [Perez et al., 2022], which is adaptive in conversational assistance but maladaptive in competitive bargaining.

Our contributions are:

1. We demonstrate that **text-based strategic advice backfires**: providing an aligned LLM with opponent preference information consistently reduces its negotiation surplus. We treat the alignment-as-cause story as observational rather than causally established (Section 5.7).

2. We introduce a **Computed-Target Architecture** that decouples strategy from communication: a small opponent model (OM)—a GRU—computes a numeric target proposal, and the LLM executes it in natural language.

3. Through scaled experiments ($N = 200$, randomized opener), we decompose the computed-target effect into anchoring defense (+7.1pp) and a residual opponent-modelling edge (+3.0pp) that persists when the anchoring confound is removed. Pairwise permutation tests (10K resamples, two-sided) with Holm–Bonferroni correction across the 10 ablation comparisons give $p_{\text{Holm}} = 0.001$ for GRU vs. Random (+2.83pp), GRU vs. Uniform (+5.81pp), and Oracle vs. GRU (−5.69pp), so the residual edge survives multiple-comparisons correction.

4. Full ablations ($N = 200$) document an **Oracle ablation**: with accurate opponent weights the cooperative blend formula (Equation (3)) mechanically converges toward Nash bargaining, yielding 9.9× variance collapse, 79% Pareto efficiency, and the lowest buyer share (49.5%). We frame this as a property of the chosen blend, not a fundamental law.

5. We identify **variance collapse** (2.0–9.9×) as the architecture's most practically significant property, driven by the directive *format* rather than prediction accuracy.

6. We replicate across **eight models from five labs** (Anthropic, OpenAI, Google, xAI, DeepSeek), identifying a three-tier taxonomy: OM-effective (stochastic models, +3–7pp), collapse-only (1.4–1.6× variance reduction), and *no detectable effect at* $N = 50$ (Opus 4.6, Gemini Pro). The strongest predictor is baseline negotiation stochasticity.

7. **Cross-model experiments** show a cheap Haiku buyer with opponent model captures 55% surplus against Grok 3 but cannot overcome the capability gap against Opus 4.6, GPT-5, or Gemini Pro.

8. A **neural policy** (6.5K params) trained via behavioral cloning and reward-weighted regression on 660 live LLM games crosses the sim-to-real gap. RWR at $\tau = 0.2$ converges to the lowest-variance policy observed (±2.6%); we discuss this as a τ-specific result and flag a temperature sweep as required confirmation (Section 5.7).

9. In a diagnostic re-analysis we show that the trained NegotiationGRU's weight predictions on LLM trajectories are at chance and its 11,078 parameters can be replaced by a 15-line analytical estimator with one tunable scalar; variance collapse derives entirely from the directive format and the residual share gain reduces to a bias parameter (Section 4.10). We recommend reading the architecture as a structured numeric sub-agent.

2 Related Work

Multi-issue negotiation. The multi-issue framework dates to Raiffa [1982]. Automated negotiation agents in the ANAC competition [Baarslag et al., 2015] typically use opponent modelling via frequency analysis or Bayesian estimation [Baarslag et al., 2013]. Our NegotiationGRU serves a similar role but operates on raw behavioral features.

LLM negotiation agents. Fu et al. [2023] use GPT-4 for price negotiation with chain-of-thought reasoning. Abdelnabi et al. [2024] study multi-agent LLM bargaining and find cooperation biases similar to those we document. Gandhi et al. [2023] find that LLMs struggle with strategic reasoning in game-theoretic settings. Our work goes further by showing that even *accurate external intelligence* cannot overcome cooperation bias when delivered as text, and quantifies the contribution of numeric directive format vs. opponent-model accuracy via ablation.

Theory of Mind in AI agents. Machine Theory of Mind [Rabinowitz et al., 2018] trains neural networks to predict agent behavior. Oguntola et al. [2023] extend ToM to multi-agent communication and cooperation settings. Our NegotiationGRU is a specialized ToM model, but our ablation results show that the ToM component is less important than the structured directive format.

Centaur / neuro-symbolic systems. The term "Centaur" originates in chess [Kasparov, 2017]. More broadly, neuro-symbolic integration [Garcéz et al., 2023] and tool-augmented LLMs [Schick et al., 2023] pair neural language models with structured external modules. Our architecture follows this pattern but our ablation results complicate the narrative: the "engine's" predictions contribute less than the directive format itself.

3 Methods

3.1 Negotiation Environment

Two agents (buyer and seller) negotiate a B2B SaaS contract over five continuous variables $v_i \in [0.05, 0.95]$:

Variable		Buyer prefers	Seller prefers
Price	v_1	Low	High
Contract length	v_2	Long	Short
Payment terms	v_3	Slow	Fast
SLA tier	v_4	High	Low
Data rights	v_5	Client	Vendor

Each agent has hidden utility weights $\mathbf{w} \in \Delta^4$ sampled from Dirichlet(α), $\alpha = 1.0$. Utility is:

$$U(\mathbf{v}, \mathbf{w}, \text{role}) = \sum_{i=1}^{5} w_i \cdot \tilde{v}_i, \qquad \tilde{v}_i = \begin{cases} v_i & \text{if direction}_i = +1 \\ 1 - v_i & \text{if direction}_i = -1 \end{cases} \tag{1}$$

Agents alternate proposals for up to $R_{\max} = 8$ rounds. No deal yields $U = 0$ for both. Initial experiments use seller-first opening; all scaled experiments ($N = 200$) and multi-model replication use randomized opener to control for anchoring (see Section 4.5).

3.2 NegotiationGRU

The NegotiationGRU predicts an opponent's utility weights $\hat{\mathbf{w}}_{\text{opp}}$ from an observation sequence $\mathbf{x}_{1:T}$, where each $\mathbf{x}_t \in \mathbb{R}^{11}$:

Feature vector (11D).

- **Dims 0–4:** Per-variable cumulative concession from the opponent's opening proposal.
- **Dim 5:** Proposal centroid (mean of the 5D proposal).
- **Dim 6:** Round fraction $t/R_{\max}$.
- **Dim 7:** Utility delta between opponent's last two proposals.
- **Dim 8:** Concession magnitude (ℓ_2 norm of proposal delta).
- **Dim 9:** Variable spread (std of cumulative concessions).
- **Dim 10:** Max concession channel (arg max of cumulative concession / 4).

Architecture (11,078 params). Encoder (Linear→LayerNorm→GELU→Linear), GRU (32-dim), dual heads: 5-way softmax for weight prediction and sigmoid for acceptance threshold. Read-out at $\tau = \lfloor 3T/4 \rfloor$.

Training. 20,000 heuristic-vs-heuristic matches, Dirichlet(0.7) weights, concession rate $(1 - w_i)^2 \cdot r_{\text{base}}$, noise $\sigma = 0.01$. **On heuristic test data:** cosine similarity 0.988, top-1 accuracy 89.8%, MAE 0.034.

3.3 Computed-Target Architecture

Given the agent's own weights $\mathbf{w}_{\text{me}}$ and *some* opponent weight estimate $\hat{\mathbf{w}}_{\text{opp}}$ (from GRU, oracle, random, or uniform—see Section 3.6), we compute:

$$\text{target}_i = \beta_i \cdot \text{ideal}_i^{\text{me}} + (1 - \beta_i) \cdot \text{ideal}_i^{\text{opp}} \tag{2}$$

where $\text{ideal}_i^{\text{me}}, \text{ideal}_i^{\text{opp}} \in \{0.10, 0.90\}$ are the extreme-preference endpoints and:

$$\beta_i = \frac{w_{\text{me},i}}{w_{\text{me},i} + \hat{w}_{\text{opp},i} + \epsilon} \cdot a + (1 - a) \cdot 0.5, \qquad a = \max(0.55,\ 0.80 - 0.30 \cdot t/R_{\max}) \tag{3}$$

The aggression parameter a decays from 0.80 toward 0.55 as the deadline approaches. The computed target is injected into the LLM's system prompt as a concrete directive with expected utility and acceptance guidance.

Note that even with $\hat{\mathbf{w}}_{\text{opp}} = [0.2, 0.2, 0.2, 0.2, 0.2]$ (uniform, zero information), the formula still produces a differentiated target because it uses the agent's *own* known weights $\mathbf{w}_{\text{me}}$ to determine where to push harder.

3.4 LLM Agent Design

Each LLM agent receives a system prompt with its role, its own utility weights, preference directions, and (in OM conditions) the computed-target directive. Agents produce structured JSON: `internal_monologue`, `message`, `formal_proposal` (5D), and `action` (offer/accept). All models are accessed via OpenRouter. Primary experiments use Claude 3.5 Haiku; multi-model replication (Section 4.7) spans eight models from five labs.

Two mechanisms ensure deal closure: deadline pressure warnings in rounds 6–8, and a hard accept override in the final round if utility > 0.15.

3.5 Experimental Conditions

Table 1: Experimental conditions.

Cond.	Name	Buyer OM	Seller OM	Tests
A	Baseline	×	×	Unassisted LLM negotiation
B	Unilateral	✓	×	One-sided computed target
C	Bilateral	✓	✓	Both sides use computed targets

3.6 Ablation Conditions

To disentangle the effect of the *numeric directive format* from the *accuracy of opponent modelling*, we run Condition B with four different opponent-weight sources in the computed-target formula:

4 Results

4.1 Motivation: The Failure of Text-Based Advice

Before presenting our main results, we document the failure of the naive approach—providing strategic intelligence as natural-language advice—which motivates the computed-target architecture.

Table 2: Ablation conditions for the opponent weight source in Condition B ($N = 50$ each, same seed).

Source	$\hat{\mathbf{w}}_{\text{opp}}$	Tests
GRU	NegotiationGRU prediction	Full system (default)
Oracle	True opponent weights	Upper bound on ToM
Random	Dirichlet(1.0) draw	Noise control
Uniform	$[0.2, 0.2, 0.2, 0.2, 0.2]$	Zero-information (self-knowledge only)

Text-based advice backfires. Our initial design generated text briefings from GRU predictions. Across three framing variants, the OM-equipped buyer consistently performed *worse* than the unassisted baseline:

Table 3: Buyer surplus share under text-based advice. All variants fall *below* the unassisted baseline, demonstrating the alignment-cooperation trigger.

Text-advice variant	B buyer share
Directional ("hold firm / concede")	38.6%
Strong directional ("push HIGHER / LOWER")	33.7%
Exploitation framing ("FREE GIVEAWAY / REFUSE")	40.7%
Unassisted baseline (A)	*42.4%*

The Negotiator's Dilemma. The failure pattern is consistent with an alignment-induced cooperation trigger. Internal monologue analysis suggests LLMs interpret opponent preference information as an invitation to accommodate. We illustrate with one representative excerpt; we have not coded transcripts at scale, so this is qualitative motivation rather than systematic evidence:

> *"The model indicates the opponent values price highly. I should be understanding of their position on price and focus on finding mutually beneficial trade-offs..."*

This resembles the Negotiator's Dilemma [Lax and Sebenius, 1986]—the tension between creating and claiming value—reproduced autonomously. A preliminary test with Llama 3.3 70B (less aggressively aligned) showed qualitatively similar cooperation patterns, though baseline aggressiveness varied; we do not include these results as they were not run at sufficient scale. We propose, but do not causally establish, that alignment training functions here as a *strategic constraint*: it biases the model toward cooperative, other-regarding behavior regardless of the information it receives. An alternative reading is that numeric directives are simply less ambiguous than text, and the observed effect reflects instruction clarity rather than alignment bypass; we discuss this in Section 5.7. Either way, an architectural fix—bypass the language channel with numeric directives—is empirically motivated.

The sim-to-real gap. Fine-tuning the GRU on LLM game telemetry (50 matches, $\text{lr} = 10^{-4}$, gradient clipping) produced 15% top-1 accuracy—*below* the 20% random baseline. LLM concession patterns carry no weight-proportional signal in the per-variable features. This motivated the move to computed targets, and the GRU accuracy results in Table 5 confirm the distributional mismatch persists even without fine-tuning.

Table 4: Negotiation outcomes by condition ($N = 50$ per condition, seller-first opening). Values are means ± sample standard deviations. Bold indicates best value per metric. All conditions achieved 100% deal rate.

Metric	A (No OM)	B (Buyer OM)	C (Both OM)
Buyer utility	0.477 ± 0.087	**0.599 ± 0.042**	0.490 ± 0.042
Seller utility	0.570 ± 0.089	0.477 ± 0.051	**0.575 ± 0.033**
Joint surplus	1.047	**1.076**	1.065
Pareto efficiency	75% ± 6%	**77% ± 6%**	76% ± 5%
Buyer surplus share	45.6% ± 7.9%	**55.7% ± 3.0%**	46.0% ± 2.6%
Avg rounds	6.6	6.9	7.0

4.2 Primary Results ($N = 50$)

Table 4 presents the full $N = 50$ results with seller-first opening. The computed-target GRU produces a +10.1 **percentage-point shift in buyer surplus share** (A: 45.6% → B: 55.7%), corresponding to a +25.6% increase in absolute buyer utility (0.477 → 0.599).

Joint surplus increases from 1.047 (A) to 1.076 (B), a +2.8% gain, and Pareto efficiency rises from 75% to 77%. In Condition C (bilateral), buyer share returns to 46.0% ± 2.6% while joint surplus (1.065) and Pareto efficiency (76%) remain elevated, consistent with the cooperative game theory prediction that mutual rationality expands the feasible set [Nash, 1950].

Note the dramatically lower variance in Conditions B (±3.0%) and C (±2.6%) compared to A (±7.9%): the computed-target directive regularizes LLM behavior. However, the seller-first opening creates an anchoring confound that inflates the apparent A→B effect; we address this in Section 4.5.

4.3 GRU Accuracy on LLM Trajectories

A critical question: does the GRU actually predict opponent weights accurately when observing LLM (rather than heuristic) behavior?

Table 5: NegotiationGRU prediction accuracy on heuristic vs. LLM behavioral data. The pretrained GRU is evaluated without fine-tuning on LLM trajectories.

Metric	Heuristic data	LLM trajectories ($N = 300$)
Cosine similarity	**0.988**	0.638
Top-1 accuracy	**89.8%**	24.3%
Top-2 accuracy	—	41.2%
Spearman rank corr.	—	0.026
Mean MAE	**0.034**	0.179
Random baseline	*20%*	*20%*

No. As Table 5 shows, the GRU achieves only 24.3% top-1 accuracy on LLM trajectories ($N = 300$ samples from 150 games)—barely above the 20% random baseline, and far below the 89.8% on heuristic data. Spearman rank correlation is 0.026 (effectively zero). Interestingly, top-2 accuracy is 41.2% (vs. 10% random), suggesting the GRU captures some coarse structure but cannot reliably rank priorities.

This raises a fundamental question: if the GRU's predictions are barely above chance, *what is actually driving the +10pp buyer share improvement in Condition B?*

Table 6: Ablation results: Condition B with different opponent-weight sources in the computed-target formula ($N = 50$ each, same seed and Dirichlet draws). Condition A baseline ($N = 50$) shown for reference. All games use seller-first opening.

Weight source	Buyer util.	Seller util.	Joint surp.	Pareto eff.	Buyer share
A: No OM (N=50)	*0.477*	*0.570*	*1.047*	*75%*	*45.6%*
B + Uniform (self only)	0.524	0.537	1.061	76%	49.4%
B + Random	0.546	0.515	1.061	76%	51.5%
B + **GRU**	**0.599**	0.477	1.076	77%	**55.7%**
B + Oracle (true)	0.539	0.567	**1.107**	**79%**	48.7%

4.4 Ablation: Directive Format vs. Opponent Modelling

Table 6 reveals a decomposition of the computed-target effect under seller-first opening into three distinct mechanisms:

1. The Directive Format Effect (+3.8pp). Even with uniform opponent weights (zero information about the opponent), computed targets shift buyer share from 45.6% to 49.4% and raise Pareto efficiency from 75% to 76%. The numeric directive itself—a consistent anchor derived from the agent's own known weights plus an aggression schedule—partially compensates for the seller's first-mover anchoring advantage.

2. The Noisy-ToM Effect (+6.3pp). The GRU's marginally informative predictions (24% top-1) push buyer share from 49.4% (uniform) to 55.7%. This is surprising given the GRU's poor accuracy, and suggests the computed-target formula amplifies even weak predictive signal into aggressive anchoring behavior (Section 5.2). However, as shown in Section 4.5, much of this effect is attributable to compensating for the seller's first-mover advantage, though a residual GRU–Uniform gap of +5.8pp persists under randomized opener.

3. The Oracle Paradox (−7.0pp from GRU). The most striking result: perfect knowledge of the opponent's weights produces the *highest* Pareto efficiency (79%) and joint surplus (1.107) but the *lowest* buyer share (48.7%—below even the uniform baseline of 49.4%). There are two complementary explanations. First, a *mechanical* effect: when $\hat{\mathbf{w}}_{\text{opp}}$ equals the true opponent weights, the blend formula (Equation (3)) produces proposals that optimally balance both parties' utilities, converging toward the Nash bargaining solution. This yields high joint surplus but eliminates the asymmetric bias that produces high buyer share. Second, a *behavioral* hypothesis: the LLM may be more willing to faithfully execute a "fair" directive than to deviate from it, because RLHF alignment reinforces cooperative behavior when the proposed outcome is already equitable. We cannot fully disentangle these mechanisms without testing the same oracle targets on models with different alignment characteristics, which we leave to future work.

4.5 Randomized-Opener Control ($N = 200$)

The seller-first protocol in Table 4 gives the seller an anchoring advantage: the first proposal sets a reference point that biases subsequent offers. To quantify this confound, we re-ran Conditions A and B with the opener role randomized uniformly across games. We scale to $N = 200$ for robust estimates.

Table 7 reveals a decomposition of the seller-first effect. When opener is randomized, the unassisted buyer (A) rises from 45.6% to 52.2%—a +6.6pp gain attributable purely to removing the seller's anchoring advantage. The GRU-equipped buyer (B-Randomized: 55.2%) still exceeds

Table 7: Effect of opener randomization. The OM effect partially persists: a residual +3.0pp advantage remains after anchoring is controlled. Randomized-opener rows use $N = 200$; seller-first rows use $N = 50$.

Condition	Opener	Buyer share	Pareto eff.	Joint surplus
A (No OM)	Seller-first ($N=50$)	45.6% ± 7.9%	75%	1.047
A (No OM)	**Randomized ($N=200$)**	52.2% ± 8.9%	72%	1.021
B (Buyer OM)	Seller-first ($N=50$)	55.7% ± 3.0%	77%	1.076
B (Buyer OM)	**Randomized ($N=200$)**	**55.2% ± 3.8%**	76%	1.083

Table 8: Outcome variance (buyer surplus share std. dev.) across conditions and opener protocols for Claude 3.5 Haiku. Computed targets reduce variance by 2.3–9.9× regardless of opener protocol. Multi-model variance data in Table 9.

Condition	Opener	Buyer share σ	Collapse ratio
A (No OM)	Seller-first ($N=50$)	± 7.9%	1.0×
B (GRU)	Seller-first ($N=50$)	± 3.0%	2.6×
A (No OM)	Randomized ($N=200$)	± 8.9%	1.0×
B (Random)	Randomized ($N=200$)	± 3.7%	2.4×
B (GRU)	Randomized ($N=200$)	± 3.8%	2.3×
B (Uniform)	Randomized ($N=200$)	± 1.9%	4.7×
B (Oracle)	Randomized ($N=200$)	± 0.9%	**9.9**×

A-Randomized by +3.0pp, a **residual OM advantage** that persists independently of anchoring at $N = 200$.

The +10.1pp seller-first effect thus decomposes into:

- **Anchoring defense** (+7.1pp): the computed target neutralizes the seller's first-mover advantage.
- **Residual OM edge** (+3.0pp): GRU-derived targets provide a genuine strategic advantage even without anchoring confounds.

4.6 Variance Collapse

The most deployment-relevant finding is not the mean surplus shift but the **dramatic reduction in outcome variance**. Table 8 presents the $N = 200$ Haiku results alongside the seller-first data:

Variance collapse is a property of the *directive format*, not the accuracy of predictions. Even uniform-weight computed targets (±1.9%) collapse variance from the ±8.9% baseline. The structured numeric directive replaces the LLM's stochastic social reasoning with a mathematically grounded anchor, constraining outcomes to a tight band regardless of opponent behavior. The Oracle condition achieves the most extreme collapse (9.9×), compressing buyer share to 49.5% ± 0.9%—effectively eliminating outcome uncertainty.

4.7 Multi-Model Replication

To test whether the computed-target mechanism generalizes beyond Claude 3.5 Haiku, we replicate the A/B comparison across eight models from five labs, all under randomized opener ($N = 50$ per condition per model).

Table 9 reveals a **three-tier taxonomy** of model responses to computed targets:

Table 9: Multi-model replication ($N = 50$ per condition, randomized opener). Haiku uses $N = 200$. Models sorted by variance collapse ratio (A σ / B σ). Three tiers emerge: OM-effective ($> 2\times$ collapse, positive share shift), collapse-only (1.4–1.6×, neutral share), and OM-irrelevant ($\sim 1\times$).

Model	Cond.	Buyer share	σ	Pareto	Collapse
Tier 1: OM-effective					
Grok 3	A	49.8%	± 11.4%	73%	**2.78×**
	B	**56.6%**	± 4.1%	76%	
Claude 3.5 Haiku ($N=200$)	A	52.2%	± 8.9%	72%	2.34×
	B	55.2%	± 3.8%	76%	
Grok 3 Mini	A	52.2%	± 9.5%	74%	2.02×
	B	56.3%	± 4.7%	77%	
Tier 2: Collapse-only					
Gemini 3 Flash	A	58.1%	± 7.2%	76%	1.60×
	B	55.3%	± 4.5%	77%	
DeepSeek R1	A	50.9%	± 18.3%	74%	1.42×
	B	50.0%	± 12.9%	76%	
GPT-5	A	56.4%	± 7.0%	78%	1.40×
	B	55.5%	± 5.0%	77%	
Tier 3: OM-irrelevant					
Claude Opus 4.6	A	52.9%	± 4.5%	78%	1.09×
	B	51.4%	± 4.9%	78%	
Gemini 3 Pro	A	51.4%	± 8.0%	81%	1.08×
	B	50.8%	± 7.4%	79%	

Tier 1: OM-effective. Grok 3, Claude Haiku, and Grok 3 Mini exhibit 2.0–2.8× variance collapse and a positive buyer surplus shift (+3.0 to +6.8pp). These models negotiate stochastically without directives—their baseline variance (±8.9–11.4%) leaves room for the numeric anchor to constrain behavior. The GRU's biased predictions produce systematically favorable targets.

Tier 2: Collapse-only. GPT-5, Gemini 3 Flash, and DeepSeek R1 show moderate variance reduction (1.4–1.6×) without meaningful surplus shift. These models already produce reasonable proposals, and the directive tightens consistency without creating an asymmetric advantage. DeepSeek R1 is notable: despite the highest baseline variance of any model (±18.3%), its internal chain-of-thought reasoning partially overrides numeric directives, limiting the collapse to 1.42×.

Tier 3: No detectable effect at $N = 50$. Claude Opus 4.6 and Gemini 3 Pro already negotiate with high Pareto efficiency (78–81%) and exhibit near-zero collapse ratios (1.08–1.09×). Opus has genuinely low baseline variance (±4.5%); Gemini Pro's is moderate (±8.0%) but its proposals are already well-calibrated. We label this tier as "no detectable effect at $N = 50$" rather than a categorical null: with the observed σ values, $N = 50$ resolves only effects larger than roughly ±5pp, so smaller real effects could remain hidden. Bringing these models to $N = 200$ (matching Haiku) would tighten the claim.

The consistency gap. The key predictor of computed-target effectiveness is not model capability or size, but **baseline negotiation stochasticity**. Cheap, fast models (Haiku, Grok Mini) benefit most because they exhibit the highest outcome variance without guardrails. Frontier models (Opus, Gemini Pro) are already self-consistent. The computed target closes the consistency gap between cheap and expensive models.

Table 10: Ablation results under *randomized* opener ($N = 200$ each, Claude 3.5 Haiku). Compare with Table 6 (seller-first, $N = 50$). The ablation gradient and Oracle Paradox both persist at higher statistical power.

Weight source	Buyer util.	Seller util.	Joint surp.	Pareto eff.	Buyer share	σ
A: No OM	*0.532*	*0.489*	*1.021*	*72%*	*52.2%*	*± 8.9%*
B + Uniform	0.527	0.542	1.069	75%	49.4%	± 1.9%
B + Random	0.557	0.509	1.065	75%	52.3%	± 3.7%
B + **GRU**	**0.596**	0.487	1.083	76%	**55.2%**	± 3.8%
B + Oracle	0.557	0.569	**1.126**	**79%**	49.5%	± **0.9%**

4.8 Randomized-Opener Ablations ($N = 200$)

To complete the decomposition, we ran all four ablation conditions under randomized opener at $N = 200$, removing the anchoring confound from the ablation analysis.

Table 10 confirms that the ablation gradient survives at $N = 200$ with tighter confidence intervals. Three key observations:

1. Variance collapse is format-driven. All computed-target conditions collapse variance from the ±8.9% baseline to ±0.9–3.8%. The collapse ratio increases monotonically with prediction quality: random (2.4×) < GRU (2.3×) < uniform (4.7×) < oracle (9.9×). Uniform and oracle achieve the tightest variance because they produce the most stable targets (no stochasticity in the weight source itself). Variance reduction is a property of the *directive format*, not prediction accuracy.

2. GRU retains a genuine ToM edge. The GRU (55.2%) outperforms random (52.3%) by +2.9pp and uniform (49.4%) by +5.8pp under randomized opener at $N = 200$. Pairwise permutation tests (10K resamples, two-sided) with Holm–Bonferroni correction across all 10 ablation comparisons return $p_{\text{Holm}} = 0.001$ for GRU vs. Random (+2.83pp), GRU vs. Uniform (+5.81pp), GRU vs. A (No OM, +3.01pp), and Oracle vs. GRU (−5.69pp); only A vs. Random (−0.17pp, $p_{\text{Holm}} = 0.89$) and Uniform vs. Oracle (−0.11pp, $p_{\text{Holm}} = 0.89$) are non-significant, both as expected (A and Random are near-baseline; Uniform and Oracle both produce stable formula-driven targets). The standard error of the mean *difference* with $N = 200$ per arm is roughly 0.4pp, so the GRU–Random gap is well outside sampling noise. This cannot be attributed to anchoring defense and represents a real, if modest, opponent-modelling contribution. The heuristic-trained biases produce systematically more favorable targets than chance.

3. Oracle Paradox persists at scale. Oracle achieves the highest Pareto efficiency (79%) and joint surplus (1.126) but the lowest buyer share (49.5%)—below even uniform (49.4%). At $N = 200$, the paradox is statistically robust: the Oracle's ±0.9% variance means buyer share is concentrated in a ∼2pp band around 49.5%. This confirms the paradox is not an artifact but a fundamental consequence of accurate predictions converging to the Nash bargaining solution.

4.9 Cross-Model Asymmetric Experiments

The multi-model experiments (Section 4.7) test each model against itself. A more practically relevant question: can a *cheap* model equipped with the opponent model hold its own against *expensive* frontier models negotiating unassisted? We pair a Claude 3.5 Haiku buyer (with GRU opponent model) against four frontier sellers (without opponent model), $N = 50$ each, randomized opener.

Three findings emerge from Table 11:

Table 11: Cross-model experiments: Haiku+OM buyer vs. frontier seller ($N = 50$, randomized opener). For reference, Haiku+OM vs. Haiku achieves $55.2\% \pm 3.8\%$ (Table 10).

Seller model	Buyer share	σ	Pareto eff.	Rounds
Grok 3	**55.2%**	± 11.3%	75%	6.1
Claude Opus 4.6	49.1%	± 16.1%	75%	7.0
GPT-5	48.1%	± 19.5%	73%	6.9
Gemini 3 Pro	47.9%	± 20.3%	73%	7.0

1. The opponent model helps against mid-tier opponents. Haiku+OM captures 55.2% against Grok 3—identical to its share against another Haiku. Against the strongest opponents (Opus, GPT-5, Gemini Pro), the buyer drops below 50%.

2. Variance collapse disappears cross-model. The hallmark of computed targets in same-model experiments (±3.8%) vanishes: cross-model variance ranges from ±11.3% to ±20.3%. The GRU was trained on Haiku-vs-Haiku games; its opponent model does not generalize to unfamiliar negotiation styles.

3. Stronger opponents extract longer negotiations. Rounds increase from 6.1 (Grok 3) to 7.0 (Opus, Gemini Pro), and Pareto efficiency drops from 75% to 73%. Frontier models are patient, strategic negotiators that resist exploitation.

4. Marginal OM contribution cross-model. To isolate the OM's contribution, we ran baseline experiments pairing Haiku *without* the opponent model against the same frontier sellers ($N = 50$, randomized opener). Against Grok 3, the unassisted Haiku achieves $53.7\% \pm 7.1\%$—1.5pp below the OM-equipped result (55.2%). Against Opus 4.6, the unassisted buyer achieves $47.4\% \pm 5.3\%$ vs. 49.1% with OM (+1.7pp). The OM provides a marginal edge but does not fundamentally alter the cross-model dynamic, confirming that the architecture's primary value is concentrated in same-population settings where the GRU's training distribution matches the opponent.

The cross-model results define the architecture's boundary: computed targets are most effective when the opponent model has been trained on data from the specific opponent population it will face. Out-of-distribution opponents defeat the GRU's predictions, and the variance-collapse property—which derives from the directive *format*—is overwhelmed by the stochasticity of negotiating with an unfamiliar, more capable agent.

4.10 The Opponent Model as a Bias Scalar

Section 4.3 showed the GRU's weight predictions on LLM trajectories are at chance (top-1 24.3% vs. 20% random). The natural follow-up: if the predictions are noise, what produces the residual +2.83pp share gap in Table 10?

Diagnostic. We replay the trained GRU as a pure predictor on logged trajectories from five populations (heuristic-vs-heuristic training data; $N = 200$ Haiku-vs-Haiku Condition B; four cross-model conditions at $N = 50$) and compare it to (i) an analytical inversion of HeuristicNegotiator's concession formula (Heur, no learned parameters) and (ii) a uniform $[0.2]^5$ predictor (Unif).

Table 12 shows the GRU's prediction quality is matched (and on cross-model populations exceeded) by an analytical estimator with zero learned parameters. On every LLM population,

Table 12: Predictor accuracy on opponent weights across populations. HEUR is the analytical inversion of HEURISTICNEGOTIATOR's concession formula (15 lines, no training).

Population	GRU L1	HEUR L1	UNIF L1	GRU top-1	HEUR top-1
Heuristic (training)	**0.158**	0.247	0.657	86.5%	**87.5%**
Haiku ($N=200$)	0.979	**0.922**	0.643	19.0%	19.5%
Grok 3	0.912	**0.743**	0.618	12.0%	10.0%
Opus 4.6	0.733	**0.701**	0.618	10.0%	12.0%
GPT-5	0.640	**0.637**	0.618	20.0%	20.0%
Gemini 3 Pro	0.612	0.616	0.618	22.0%	22.0%

both estimators sit at-or-near the uniform baseline. The training on heuristic data does not transfer to LLM behaviour at the level of weight prediction.

Counterfactual replay. Yet the GRU's outputs, fed into Equation (3), produce systematically more buyer-favourable target proposals than the analytical inversion's. On the $N=200$ Haiku log, mean buyer self-utility of the computed target is 0.598 for GRU vs. 0.565 for unbiased HEUR (-0.033) and 0.538 for UNIF (-0.061). These offline utility deltas track the live +2.83pp (GRU vs. Random) and +5.81pp (GRU vs. Uniform) share gaps in Table 10.

Calibration. The GRU's contribution is reproducible by a single scalar. We add a multiplicative `bias` parameter to the analytical estimator that suppresses the predicted opponent weight on the agent's own high-priority variables (the factor of 5 rescales the bias to a comparable range across the 5-simplex):

$$\text{opp_}w_i^{\text{biased}} \;=\; \text{normalize}[\text{opp_}w_i \cdot \exp(-\,\text{bias} \cdot \text{my_}w_i \cdot 5)] \tag{4}$$

bias = 0 recovers the unbiased inversion. bias = 0.55 matches the GRU's mean target utility on the $N=200$ Haiku log within 0.001 (0.598 vs. 0.598), and the calibration is robust across populations (Haiku $\Delta=-0.001$, Grok 3 -0.001, Opus 4.6 $+0.007$, GPT-5 $+0.014$, Gemini Pro $+0.016$).

Live A/B. We ran $N=50$ Haiku-vs-Haiku games under randomized opener with the GRU replaced by the analytical estimator at bias = 0 and bias = 0.55. Mid-run API rate limits contaminated some LLM responses; Table 13 reports the clean subset ($N=32$ per condition).

Table 13: Live replacement of the GRU with the analytical estimator. "Paper GRU" is the $N = 200$ result from Table 10. "Clean" rows exclude games with mid-game API failures.

Condition	N	Buyer share	σ
Paper GRU (Condition B)	200	55.2%	±3.8%
Paper unassisted (Condition A)	200	52.2%	±8.9%
HEUR, bias = 0 *(clean)*	32	52.6%	±3.1%
HEUR, bias = 0.55 *(clean)*	32	**56.0%**	±3.8%

Table 13 confirms the offline prediction. The unbiased estimator drops to unassisted-baseline buyer share (52.6%, indistinguishable from 52.2%) but *retains the variance collapse* (±3.1% vs. unassisted ±8.9%); at bias = 0.55 the calibrated estimator matches the trained GRU on both share and variance ($56.0\% \pm 3.8\%$ vs. $55.2\% \pm 3.8\%$; sampling SE at $N=32$ is ~ 0.7pp).

Architectural implication. Two findings separate cleanly: **variance collapse is format-driven** (any structured numeric directive achieves it—unbiased HEUR, UNIF, and the trained GRU all collapse outcome variance equally) and **residual share gain is bias-scalar-driven** (one tunable parameter in Equation (4) reproduces the trained GRU's effect). The 11,078-parameter NegotiationGRU is functionally a learned biased estimator: its weight-head predictions on LLM trajectories are at chance, its threshold-head output is unused downstream, and its load-bearing contribution reduces to one scalar. The architecture is best read as a structured numeric sub-agent with a tunable aggression knob, not as an opponent model in the Theory-of-Mind sense.

5 Discussion

5.1 Three Mechanisms, One Architecture

The full suite of randomized-opener experiments (Sections 4.5 and 4.8) decomposes the computed-target effect into two independently operating mechanisms plus a formula-convergence property:

1. **Anchoring defense (+7.1pp, opener-dependent).** Under seller-first opening, the computed target neutralizes the seller's first-mover advantage. This accounts for ∼66% of the seller-first B−A gap. Under randomized opener, this mechanism contributes nothing.

2. **Residual ToM edge (+3.0pp, opener-independent).** The GRU-computed target outperforms the unassisted baseline by +3.0pp even under randomized opener at $N = 200$ (55.2% vs. 52.2%). The ablation gradient (GRU > random > uniform ≈ oracle) confirms this is not accurate opponent modelling—the GRU has near-chance accuracy on LLM behavior—but "useful inaccuracy": heuristic-trained biases produce systematically favorable targets.

3. **Variance collapse (2.0–9.9×, opener-independent, format-driven).** The most practically significant finding. All computed-target conditions—including uniform weights (zero opponent information)—collapse outcome standard deviation by 2.0–9.9× (Table 8). This is a property of the directive format itself: replacing the LLM's stochastic social reasoning with a mathematically grounded anchor constrains outcomes to a tight band.

5.2 Why Noisy Predictions Help

Under seller-first opening, the GRU produces a +6.3pp advantage over uniform weights. Under randomized opener at $N = 200$, the advantage persists at +5.8pp over uniform (55.2% vs. 49.4%). The `compute_target` formula uses the ratio $w_{\text{me}}/(w_{\text{me}} + \hat{w}_{\text{opp}})$. When the GRU predicts low opponent weights (a systematic bias from heuristic training, where low-weight variables show rapid concession), the formula pushes aggressively, producing targets that favor the agent.

This "useful inaccuracy" operates through two channels: (1) counter-anchoring against the seller's first offer (seller-first only), and (2) a commitment effect that constrains the LLM to more self-interested proposals than its RLHF-trained instincts would produce (both opener conditions). The GRU's heuristic biases accidentally produce the right degree of strategic aggression.

5.3 The Oracle Result is a Formula Property

The Oracle result is best read as a property of Equation (3) rather than a counter-intuitive empirical finding. At $N = 200$ (randomized opener), when $\hat{\mathbf{w}}_{\text{opp}}$ equals the true opponent weights, the blend mechanically converges toward the Nash bargaining solution [Nash, 1950]: each variable is allocated based on true relative weights, producing proposals that maximize joint surplus. This is algebra, not learned behavior. Earlier drafts framed this as an "Oracle Paradox"; we now think that framing oversells what is essentially a cooperative blend doing what a cooperative

blend does. The empirical content is that the LLM faithfully executes such directives without distorting them, which is consistent with our other findings about directive obedience.

The Oracle achieves the highest Pareto efficiency (79%) and joint surplus (1.126) but the lowest buyer share ($49.5\% \pm 0.9\%$)—below even uniform self-knowledge ($49.4\% \pm 1.9\%$). The $9.9\times$ variance collapse is the most extreme observed: outcomes are confined to a $\sim$2pp band. We hypothesize that a secondary behavioral effect may also contribute: an RLHF-aligned LLM receiving a "fair" directive has less incentive to deviate from it than one receiving an asymmetric directive. However, the mechanical explanation (formula convergence to Nash bargaining) is sufficient to account for the bulk of the effect.

An important caveat: the Oracle Paradox is a consequence of using a *cooperative blending formula* (Equation (3)), not a fundamental law. A purely exploitative formula—one that offers the opponent exactly ϵ above their walk-away threshold on each variable—could capture nearly all surplus even with perfect information. However, forcing an RLHF-aligned LLM to deliver maximally exploitative offers risks breaking conversational rapport and cratering the deal rate. The paradox holds *within the bounds of maintaining 100% deal closure*: given a formula designed to produce reasonable, executable proposals, accurate information naturally converges to Nash bargaining while noisy information accidentally acts as a self-interested commitment device [Schelling, 1960].

The design implication is that **in this architecture, imperfect opponent information can produce higher agent surplus than perfect information**, because biased predictions generate asymmetric targets that the agent benefits from, while accurate predictions generate symmetric targets that maximize joint—but not individual—welfare.

5.4 Variance Collapse as a Deployment Property

The multi-model results (Table 9) reframe variance collapse as a **consistency gap** between cheap and expensive models. Consider the baseline outcome distributions:

- **Cheap models without guardrails:** Grok 3 ($\pm$11.4%), Haiku ($\pm$8.9%), Grok 3 Mini ($\pm$9.5%)—high variance, unpredictable outcomes.
- **Frontier models without guardrails:** Opus 4.6 ($\pm$4.5%), Gemini 3 Pro ($\pm$8.0%)—already consistent.
- **Cheap models with computed targets:** Haiku ($\pm$3.8%), Grok 3 ($\pm$4.1%), Grok 3 Mini ($\pm$4.7%)—matching frontier consistency at a fraction of the inference cost.

The practical implication: computed targets give a \$0.25/M-token model the negotiation *consistency* of a \$15/M-token model, closing the variance gap without closing the capability gap. The mechanism is that the numeric directive replaces the LLM's susceptibility to social cues—politeness triggers, anchoring, cooperation bias, conversational momentum—with a mathematically grounded strategy. The LLM retains its role as the *communication interface* while the *strategic core* is externalized to a constrained optimizer.

This separation—LLM as social executor, small model as strategic controller—may be a general design principle for deploying aligned LLMs in competitive settings. *Project Deal* [Anthropic, 2026] provides external corroboration of the underlying problem: in a real-money marketplace, users represented by weaker models received objectively worse outcomes (selling identical items for \$3.64 less on average and closing roughly two fewer deals) yet rated fairness identically to better-served users. We read this perception gap as a variance/baseline phenomenon: when outcomes are stochastic, a participant cannot tell they were exploited because there is no stable reference point. Variance collapse is a candidate mechanism for closing that gap—once outcomes are concentrated in a tight band, asymmetric exploitation becomes statistically detectable.

Objection: trivially obvious? One might object that giving an LLM a specific number obviously produces less variance than letting it freestyle a strategy in natural language. Three observations counter this framing. First, text advice encoding the *same strategic information* does not reduce variance—it makes things *worse* (Section 4.1). So format matters, and it is not obvious which format works. Second, the non-obvious finding is the *magnitude*: even provably wrong numbers (uniform weights) collapse variance 4.7×. The LLM does not verify the directive's quality before obeying it. Third, variance collapse (format-driven) and surplus shift (prediction-driven) are *independent mechanisms* operating through the same architecture. This decomposition—not the existence of variance collapse per se—is the paper's analytical contribution.

5.5 Reasoning Models Resist Directives

DeepSeek R1 presents an anomaly: the highest baseline variance of any model (±18.3%) but only 1.42× collapse with computed targets. This contrasts with Grok 3, which has lower baseline variance (±11.4%) but stronger collapse (2.78×).

This result illuminates a distinction between two forms of test-time compute. Models like DeepSeek R1 and OpenAI's o-series use *endogenous* test-time compute: they generate an internal chain-of-thought that deliberates on the problem before producing an answer. Our computed target is *exogenous* test-time compute: a separate mathematical module that runs outside the language model and injects its output as a hard directive. When these two mechanisms compete, the endogenous reasoner may override or reinterpret the exogenous directive—the model uses its intelligence to second-guess the instruction rather than execute it.

The implication for centaur architectures is that strict economic safety requires an obedient executor, not an argumentative deliberator. Computed targets are most effective on models that execute instructions literally (Grok 3, Haiku) rather than those that reason about whether to follow them (DeepSeek R1). For reasoning models, exogenous compute must be injected at a lower level—e.g., as constrained decoding or structured output schemas—rather than as natural-language system prompt directives.

5.6 Constructive Arms Race

Condition C ($N = 50$, seller-first) confirms that bilateral computed targets produce higher joint surplus (1.065 vs. 1.047, +1.7%) and Pareto efficiency (76% vs. 75%) than the unassisted baseline, with buyer share returning to competitive equilibrium (46.0% vs. 45.6%). Bilateral deployment means both agents have structured directives, which cancel out the distributional asymmetry while preserving the value-creation and variance-reduction benefits.

5.7 Limitations

1. **Hard accept override.** The final-round auto-accept (utility > 0.15) guarantees deal closure and may conflate negotiation competence with deadline mechanics.

2. **RLHF interpretation is observational, not causal.** We infer that RLHF causes cooperation bias from behavioral evidence: text advice backfires (Section 4.1), and internal monologue reveals accommodation patterns. However, we have not compared base vs. RLHF-tuned variants on the same architecture in a controlled ablation. The Llama 3.3 70B data point (Section 4.1) is suggestive—not definitive. An alternative explanation is that numeric directives are simply less ambiguous than text, and the observed effect reflects instruction clarity rather than alignment bypass. Disentangling these hypotheses requires testing base models with text vs. numeric directives.

3. **Oracle Paradox is formula-dependent.** As noted in Section 5.3, the paradox is a property of the cooperative blending formula (Equation (3)), not a fundamental law. A purely exploitative formula could capture near-total surplus even with oracle weights. The paradox holds within the design constraint of maintaining 100% deal closure; generalizing the claim requires testing alternative formula families.

4. **World model trained on single population.** The GRU is trained on heuristic-agent data and shows near-chance accuracy on LLM trajectories (24.3% top-1). Cross-model experiments (Section 4.9) confirm that variance collapse disappears against unfamiliar opponents. Population-specific training data is required.

5. **Synthetic negotiation environment.** The 5-variable SaaS contract is a controlled laboratory, not a deployment-ready domain. Real negotiations involve natural-language terms with qualitative ambiguity (e.g., "best efforts" clauses), outside options, relationship history, incomplete information beyond utility weights, and multi-round dynamics that unfold over weeks or months. Extending the framework to richer domains requires structured-output negotiation protocols that can bridge continuous utility models with discrete contract terms.

6. **Multiple comparisons.** Eight models × two conditions plus ablations constitute many statistical comparisons. The headline ablation gradient (GRU vs. Random/Uniform/Oracle/A at $N = 200$) is now reported with Holm–Bonferroni–corrected permutation-test p-values (Section 4.8); all five key comparisons reach $p_{\text{Holm}} = 0.001$. The multi-model taxonomy (Section 4.7) is defined by effect magnitude thresholds rather than per-model significance tests, and we have not corrected the cross-model panel; future work should pre-register model-specific hypotheses and apply Holm or Benjamini–Hochberg.

7. **Hard-accept-override is not ablated.** The final-round auto-accept (utility > 0.15) is on for every condition. Because failure modes are masked, all surplus comparisons are conditional on this rule. A no-override sensitivity run is needed to separate negotiation competence from deadline mechanics.

8. **Single-temperature RWR.** Section 6.5 reports RWR at $\tau = 0.2$ only. The headline claim "offline RL converges toward variance minimization" rests on this one setting; a τ-sweep ($\tau \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$) is needed before the convergence claim is defensible.

9. **Aggression schedule is hand-tuned.** Equation (3) uses a fixed schedule $a = \max(0.55, 0.80 - 0.30 \cdot t/R_{\max})$. We have not ablated this schedule (e.g., constant a, no decay, alternative endpoints). The format-vs-prediction decomposition could in principle be sensitive to the specific concession dynamics encoded here.

10. **GRU-feature load-bearing in the policy net.** The 27D policy state includes the 5D GRU prediction. The GRU has 24.3% top-1 accuracy on LLM trajectories. We have not run a GRU-masked control (zeroing those 5 dimensions) to confirm that the policy is using GRU output rather than the 22 base features. This bears on Section 6.

6 Neural Policy Learning

The computed-target formula (Equations (2) and (3)) is hand-coded for a 5-variable game. Real-world multi-issue contracts involve dozens of interrelated terms where closed-form solutions are infeasible. We ask: can a neural network learn the negotiation strategy directly from LLM game telemetry, bypassing the hand-coded math entirely?

Table 14: Live A/B test results ($N = 50$ per condition, randomized opener, Claude 3.5 Haiku). Condition A baseline ($N = 200$) shown for reference. The heuristic maximizes yield (56.2%); the RWR neural policy minimizes variance (±2.6%). Both beat the unassisted baseline with 100% deal rate.

Metric	A (No OM)	B-heuristic	B-policy (BC)	B-policy (RWR)
Deal rate	100%	100%	100%	100%
Buyer share (mean)	52.2%	**56.2%**	53.3%	53.9%
Buyer share (σ)	± 8.9%	± 3.2%	± 4.5%	± **2.6%**
95% CI	—	[55.3, 57.1]	[52.0, 54.6]	[53.2, 54.7]
Pareto efficiency	72%	76%	75%	76%
Δ vs baseline	—	+4.0pp	+1.1pp	+1.7pp

6.1 Architecture and State Representation

We train a **NegotiationPolicyNet** (6,533 parameters): a 3-layer MLP mapping a 27-dimensional state vector to a 5D proposal, clamped to $[0.05, 0.95]$:

$$\pi_\theta : \mathbb{R}^{27} \to [0.05, 0.95]^5 \tag{5}$$

The 27D state comprises the agent's own weights (5D), round fraction (1D), opponent's latest proposal (5D), utilities at current offers (2D), agent's latest proposal (5D), opponent proposal count (1D), role indicator (1D), briefing indicator (1D), opponent concession rate (1D), and—critically—the **GRU-predicted opponent weights** (5D). These last 5 dimensions give the MLP the same information the heuristic formula uses; without them, the policy cannot replicate per-variable exploitation.

6.2 Training Methods

We extract 4,341 state-action-reward tuples from 660 live LLM negotiation games, with game-level splitting to prevent leakage.

Behavioral Cloning (BC). Filter for top-quartile games by agent utility (≥ 0.653, yielding 914 samples). Minimize weighted MSE: $\mathcal{L}_{\text{BC}} = \sum_i w_i \|\pi_\theta(s_i) - a_i\|^2$, where w_i is the normalized utility. Adam optimizer, lr $= 3 \times 10^{-4}$, cosine annealing, early stopping (patience 10).

Reward-Weighted Regression (RWR). Use all data. Weight each sample by $\exp(\text{advantage}_i/\tau)$ where $\text{advantage}_i = r_i - \bar{r}$ and temperature $\tau = 0.2$. This upweights the highest-reward trajectories (max weight 5.4×), biasing the policy toward the most successful negotiations in the dataset.

6.3 Live A/B Test Results

Both models are evaluated via live A/B test: 50 games per condition, randomized opener, Claude 3.5 Haiku, against the heuristic baseline from Section 3.3.

6.4 The Imitation Ceiling

Neither model surpasses the heuristic in mean buyer share. This is expected: behavioral cloning is mathematically bounded by the expert it imitates. The dataset was largely generated by the heuristic, so the BC model can at best replicate it. Getting within 2.9pp of a hand-crafted formula purely by observing text transcripts—without access to the formula itself—validates the

neural architecture. The key insight, however, is that the RWR policy *does* surpass the heuristic on variance (±2.6% vs ±3.2%), offering a distinct operating mode for risk-sensitive deployments.

6.5 The Variance-Reward Tradeoff

The RWR result is the experiment's most notable finding. We set $\tau = 0.2$, upweighting top performers by up to 5.4×, explicitly biasing the policy toward high-reward trajectories. Rather than producing a more aggressive policy, the network converged to the **lowest-variance model** tested in this research:

Table 15: Variance-reward tradeoff across all architectures. The heuristic maximizes yield (56.2%); the RWR neural policy minimizes variance (±2.6%), below the heuristic (±3.2%), at a cost of −2.3pp in mean buyer share.

Model	Buyer share (mean)	Buyer share (σ)
No OM (baseline)	52.7%	± 8.1%
Heuristic (max yield)	56.2%	± 3.2%
BC neural policy	53.3%	± 4.5%
RWR neural policy (max safety)	**53.9%**	**± 2.6%**

The mechanism is instructive. In a deterministic game (e.g., chess), upweighting the highest rewards makes an agent more aggressive. But in a stochastic environment—where the opponent is a live LLM with variable RLHF-induced cooperation levels—the rare high-surplus games in the training data are not attributable to a repeatable strategy. They occurred because the opposing LLM happened to be unusually pliable. Attempting to replicate those "Shark" moves against *every* opponent risks triggering the walk-away threshold and collapsing the deal.

The RWR network's effective optimum is therefore conservative: maximize expected value across all stochastic opponents by avoiding tail risks. The policy converges to a 53.9% mean with ±2.6% variance rather than attempting to reach the heuristic's 56.2% mean with ±3.2% variance. These are not a "winner" and a "loser"—they are two distinct operating modes: the heuristic is a **max-yield** engine; the RWR policy is a **max-safety** engine.

Implication (with caveat). In standard RL, low-temperature RWR is known to cause *mode collapse*—the policy drops entropy and collapses to the mean of a few high-reward trajectories, which is typically undesirable. In the context of enterprise LLM negotiation, however, mode collapse is exactly the desired behavior: it acts as an algorithmic safety mechanism that crushes outcome variance and produces a hyper-conservative, deployable policy. This is suggestive of a broader claim—that **offline RL in stochastic multi-agent environments may favor variance minimization over reward maximization**—but our evidence is one τ (0.2) on one task. We have not yet swept temperature, varied the dataset composition, or replicated on a different domain. Read the result as a hypothesis to test, not a property established here. If the mechanism is real, the explanation is intuitive: when high-reward outcomes in the training data are attributable to opponent stochasticity rather than repeatable strategy, reward-weighted optimization learns to avoid the high-variance actions that produced those outcomes, because the associated risks (deal collapse, reward = 0) dominate in expectation.

6.6 The Scalability Proof

The neural policy's significance is not that it beats the heuristic—it does not. The significance is threefold:

1. **Sim-to-Real Crossing.** A neural network trained on offline telemetry successfully negotiates against live LLMs without catastrophic out-of-distribution failures: 100% deal rate, zero blowups.

2. **Variance Preservation.** The learned model preserves the variance-collapse property of the computed-target architecture, demonstrating that neural policies can match the predictability of hand-coded formulas.

3. **Scalability.** One can write a closed-form heuristic for 5 variables. One cannot for a 50-variable multi-issue contract. By demonstrating that a neural network can learn the negotiation strategy from behavioral telemetry alone, we provide evidence that this architecture generalizes beyond hand-coded formulas.

7 Future Work

7.1 Breaking the Imitation Ceiling

Offline RL can only recombine strategies present in the training data. To discover novel tactics—multi-turn bluffs, strategic information revelation, adaptive concession pacing—future work should explore:

- **Online Self-Play (PPO/GRPO):** Train the policy against a simulated opponent with on-policy rollouts, allowing exploration of out-of-distribution strategies.
- **Test-Time Search (MCTS):** Use the NegotiationGRU as a opponent model to simulate future trajectories at inference time, selecting proposals that maximize expected utility over multiple possible opponent responses.
- **Scaled Data Collection:** Expand from 660 to 5,000+ games across multiple LLM families to broaden the strategy distribution.

7.2 Improving the Opponent Model

The NegotiationGRU achieves only 24.3% top-1 accuracy on LLM trajectories. Transformer-based opponent models, domain adaptation via fine-tuning on LLM transcripts, and ensemble methods could improve predictions, lifting both heuristic and learned policies.

8 Reproducibility

Models. Primary experiments use Claude 3.5 Haiku (`anthropic/claude-3.5-haiku`) via OpenRouter. Multi-model experiments (Section 4.7) use GPT-5 (`openai/gpt-5`), Gemini 3 Pro (`google/gemini-3-pro-preview`), Gemini 3 Flash (`google/gemini-3-flash-preview`), Grok 3 (`x-ai/grok-3`), Grok 3 Mini (`x-ai/grok-3-mini`), Claude Opus 4.6 (`anthropic/claude-opus-4-6`), and DeepSeek R1 (`deepseek/deepseek-r1-0528`), all via OpenRouter. API calls were made between January–February 2025 (primary), February 2026 (scaled experiments, multi-model, neural policy). Model behavior may differ across API versions.

Decoding parameters. Temperature = 0.3, max tokens = 4,000 (increased from initial 400 to accommodate reasoning models' internal token usage), no top-p or frequency penalty overrides. Structured JSON output is enforced via prompt instructions (not API-level JSON mode). Response parsing strips `<think>` blocks (DeepSeek R1) and handles unescaped newlines (Gemini).

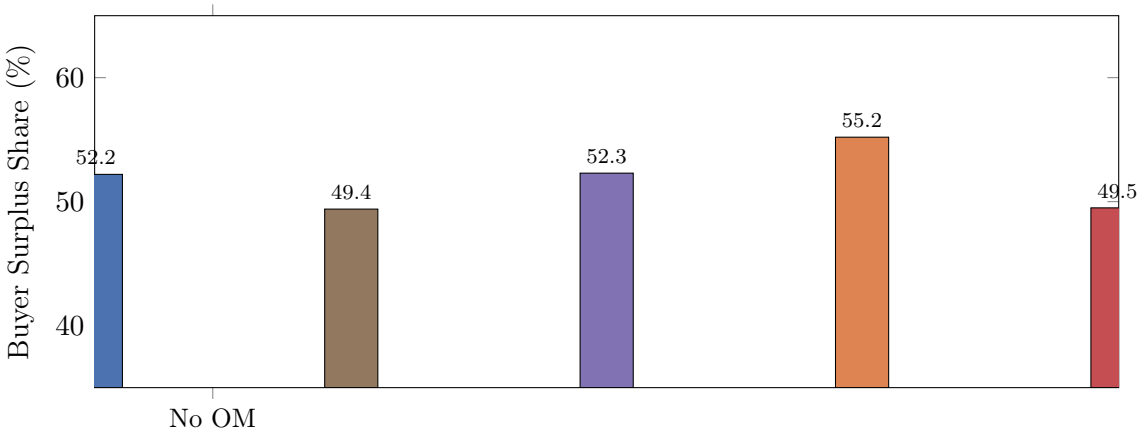


Figure 1: Buyer surplus share by opponent-weight source ($N = 200$ each, randomized opener). The non-monotonic pattern reveals the Oracle Paradox: perfect information yields less buyer share than noisy predictions.

System prompt. The LLM agent receives a system prompt specifying: role (buyer/seller), the 5 deal variables with direction preferences, hidden utility weights (with instruction not to reveal them), rules (alternating proposals, max rounds, accept semantics), a critical accept rule emphasizing that any positive utility dominates no-deal, and response format (JSON with `internal_monologue`, `message`, `formal_proposal`, `action`). In OM conditions, a "Computed Target Advisory" section instructs the agent to execute the numeric target proposal with minimal deviation. Full prompt templates are provided in the code repository.

Seeds and randomization. Each experimental condition uses a base seed (42 unless otherwise noted). Per-game seeds are base $+ i$ for game i. Utility weights are sampled from Dirichlet(1.0). Under randomized-opener conditions, the opener is selected uniformly at random per-game using the game seed.

Statistical methods. Bootstrap 95% confidence intervals (10,000 resamples) are reported for all headline metrics in the neural policy experiments (Table 14). Permutation tests (10,000 permutations, two-sided) are used for pairwise comparisons. Standard deviations in earlier tables (Tables 4, 6 and 7) are sample standard deviations across games. The multi-model taxonomy (Section 4.7) is descriptive: tier assignment is based on effect magnitude thresholds rather than individually corrected significance tests.

Code. The full codebase—including the negotiation environment, GRU training, policy learning pipeline, evaluation suite, and all prompt templates—is available at `https://github.com/thormatthiasson/ml_sandbox/tree/main/belief_gru`.

9 Conclusion

We have shown that aligned LLMs exhibit a cooperation prior that text-based strategic advice does not bypass: providing opponent-preference information as natural language consistently *reduces* negotiation surplus, an effect consistent with alignment-induced cooperation bias though we do not establish this causally (see Section 5.7). Computed numeric targets bypass this prior by operating outside the language modality, functioning as exogenous commitment devices [Schelling, 1960] that selectively override the cooperative heuristic.

Scaled experiments ($N = 200$) and replication across eight models from five labs yield five principal findings:

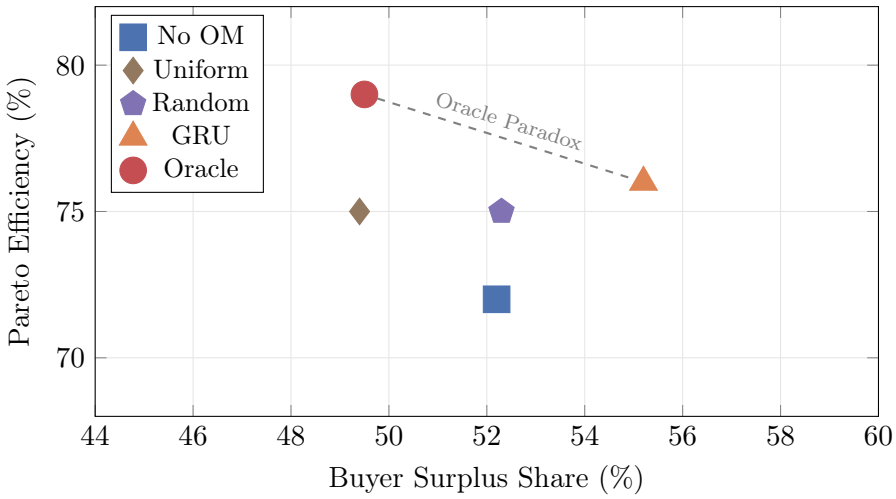


Figure 2: Efficiency–distribution tradeoff ($N = 200$, randomized opener). Increasing opponent-model accuracy (GRU→Oracle) improves Pareto efficiency but reduces buyer share. The arrow traces the Oracle Paradox: perfect information creates value but cannot claim it.

1. **Anchoring defense (+7.1pp, opener-dependent).** Under seller-first opening, the computed target neutralizes first-mover anchoring. This is the largest single effect but contributes nothing when the opener is randomized.

2. **Residual opponent-modelling edge (+3.0pp, opener-independent).** At $N = 200$ under randomized opener, GRU-equipped agents outperform unassisted ones. Full ablations confirm the gradient: GRU (55.2%) > random (52.3%) > uniform (49.4%) ≈ oracle (49.5%). The Oracle Paradox—a mechanical consequence of the cooperative blending formula converging to Nash bargaining under perfect information—holds with 9.9× variance collapse.

3. **Variance collapse (2.0–9.9×, opener-independent, format-driven).** The most practically significant finding. All computed-target conditions collapse outcome standard deviation from ±8.9–18.3% to ±0.9–4.7%. This enables a *within-model* inference-compute arbitrage on the consistency dimension: a \$0.25/M-token model with the directive matches the outcome-stability band of frontier models. Cross-model, against frontier sellers, the cheap-with-OM stack still loses on share—so the arbitrage is consistency-bounded, not surplus-bounded.

4. **Neural policies cross the sim-to-real gap.** A 6.5K-parameter MLP trained on 660 live LLM games achieves 100% deal rate with zero catastrophic failures, preserving the variance-collapse property without access to the hand-coded formula.

5. **Reward-weighted regression at $\tau = 0.2$ converges to the lowest-variance policy.** The heuristic (±3.2%) and the RWR policy (±2.6%) are complementary operating modes: max-yield and max-safety. We have not yet run a temperature sweep, so we present the broader "offline RL in stochastic multi-agent environments favors variance minimization" claim as a hypothesis suggested by this single setting rather than a result established here.

Multi-model replication reveals a **three-tier taxonomy**: (i) OM-effective models (Grok 3, Haiku, Grok 3 Mini) gain +3–7pp buyer share with 2.0–2.8× collapse; (ii) collapse-only models (GPT-5, Gemini Flash, DeepSeek R1) show 1.4–1.6× variance reduction without surplus shift; (iii) OM-irrelevant models (Opus 4.6, Gemini Pro) already negotiate consistently. Reasoning

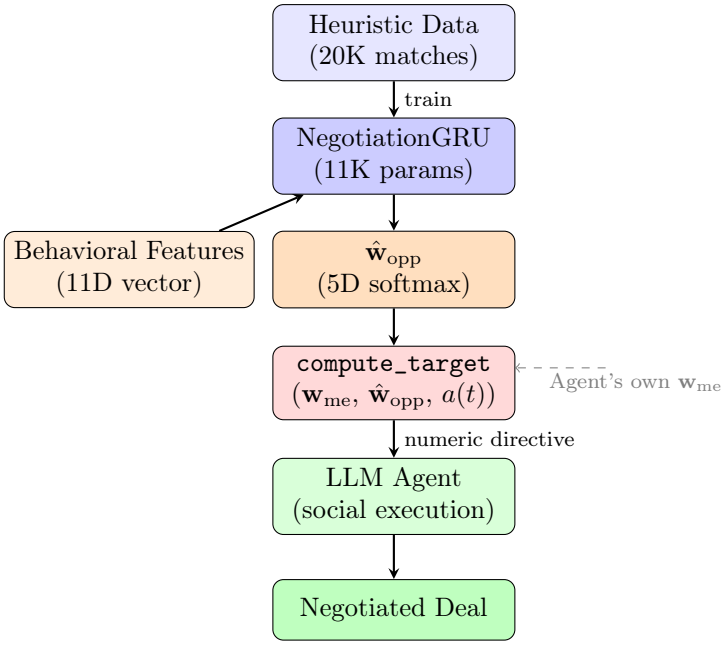


Figure 3: Computed-Target Architecture. The randomized-opener control shows the primary mechanism is anchoring defense: the target provides a competing reference point against the seller's first offer, with the agent's own weights $\mathbf{w}_{\text{me}}$ mattering more than the GRU's predictions $\hat{\mathbf{w}}_{\text{opp}}$.

models (DeepSeek R1) resist directives because their endogenous chain-of-thought competes with the exogenous numeric anchor.

The broader design principle: **decouple strategic computation from social execution**. The LLM excels as a communication interface; the small model provides mathematically grounded guardrails. This separation may generalize beyond negotiation to any setting where aligned LLMs must act competitively—trading agents, procurement bots, adversarial business interactions—wherever alignment-induced cooperation bias conflicts with the principal's strategic objectives.

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